

SQL Aggregate Functions and operators, Exercise 2 SQL

01. Students

Q1 SELECT DISTINCT department
FROM Students;

DEPARTMENT
IT
HR
FINANCE

Q2 SELECT department, AVG (age) AS avg-age
FROM Students
GROUP BY department;

Department	avg-age
IT	21.0
HR	22.0
FINANCE	23.0

Q3 SELECT department, count(*) AS student
FROM Students
GROUP BY department.
HAVING COUNT (*) > 1;

Department	Student-count
IT	2
HR	2

④ SELECT Student-id, name
age,
department

FROM Students.

WHERE age BETWEEN 21 AND 23.

Student-id	name	age	department
2	Bob	22	HR.
3	Charlie	21	IT
4	Diana	23	FINANCE
5	Eve	22	HR

⑤ SELECT Student-id

name

age

department.

FROM Students

WHERE department IN ('IT', 'HR')

AND age > 21;

Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

03 (06) SELECT department, SUM (credits)
AS total_credits
FROM courses
GROUP BY department
HAVING SUM (credits) > 5;

department	total_credits
IT	11

(07) SELECT course_id,
course_name,
department
FROM courses
WHERE credits <= 4;

course_id	course_name	department
101	SQL Basics	IT
104	Excel	Finance
105	Statistics	HR

(08) SELECT course_id
course_name,
credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

Course_id	Course name	credits
102	Python	4
103	Data Science	4
101	SAL BASICS	3

Q3 THE ENROLLMENT TABLE

⑨ SELECT MAX(grade) AS max-grade,
 MIN(grade) AS min-grade,
 Avg(grade) AS avg-grade
 FROM enrollments;

Max_grade	Min-grade	avg-grade
90	78	84.6

⑩ SELECT course_id, count(*) AS enrollment
 FROM enrollments
 GROUP BY course_id

Course_id	Enrollment-count
101	1
102	1
103	1
104	1
105	1

04 THE SALARIES TABLE

- ⑪ SELECT department
SUM (salary) AS total_salary
SUM (bonus) AS total_bonus
FROM salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

- ⑫ SELECT department,
AVG (salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary) > 55000;

department	avg_salary
IT	61000
Finance	70000

- ⑬ SELECT employee_id,
Salary,
bonus, (salary + bonus) AS total
compensation
FROM salaries
WHERE (salary + bonus) > 60000;

employee id	name	Salary	bonus
1	Tom	60000	5000
3	Spice	70000	6000
4	Tyke	62000	8500

Q5 THE PROJECT TABLE

- ⑭ SELECT department,
 SUM (budget) AS total - budget,
~~AVG~~ (budget) AS avg - budget.
 FROM projects
 GROUP BY department
 HAVING AVG (budget) > 70000;

department	total - budget	avg - budget
IT	270000	135000

- ⑮ select project_id,
 project_name,
 department,
 budget
 FROM projects
 WHERE budget BETWEEN 50000 AND 120000
 AND department <> 'Marketing';

Project_id	project-name	department
1	AI App	IT
3	Payroll System	Finance
5	HR Portal	HR